

PROFESSIONAL SUMMARY

Data Scientist with 2+ years of experience building AI/ML and LLM-based systems at scale. Specialized in prompt engineering, NLP, customer segmentation, and conversational AI (Text-to-SQL) using Databricks, Snowflake, and Python. Proven track record in delivering measurable business impact , including a 95% + reduction in manual effort and a 30% increase in user conversion through end-to-end data pipelines, LLM classification systems and self-serve analytics platforms.

EDUCATION

Year	Degree/Certificate	Institute	CPI/%
2022–2024	M.Tech / Computer Science & Engineering	Indian Institute of Technology, Kanpur	8.1/10.0
2019–2022	B.Tech / Computer Science & Engineering	Charotar University of Science & Technology, Anand	8.7/10.0
2016–2019	Diploma / Computer Engineering	A.V. Parekh Technical Institute, Rajkot	8.7/10.0
2016	GSEB (Class X)	Sadhu Vaswani School, Rajkot	80%

EXPERIENCE

- **Interglobe Aviation (IndiGo)**
Data Scientist

June 2024 – Present
Gurugram, India

- **Intelligent Feedback Analysis System using LLMs**
 - * Engineered an incremental data ingestion pipeline from Snowflake to Databricks with daily batch jobs, handling end-to-end data cleaning and preprocessing within a production-grade pipeline.
 - * Designed few-shot prompt templates to classify customer feedback into 3 sentiment classes (Positive, Neutral, Developmental) and a multi-level issue taxonomy (L1→L2→L3→L4), supporting multi-label classification where a single feedback maps to multiple categories simultaneously.
 - * Conducted multi-model evaluation across Claude, LLaMA, and Mistral to benchmark classification accuracy and semantic consistency, selecting the best-performing model for production deployment.
 - * Operationalized the end-to-end NLP classification pipeline on Databricks Workflows with automated daily inference, writing structured sentiment and category outputs back to Snowflake via Delta Lake for downstream consumption.
 - * Delivered a Power BI dashboard enabling business users to explore sentiment trends, category breakdowns, and dissatisfaction signals — reducing manual feedback review effort by **95%+**.
 - **Hotel Booking Propensity Model using Customer Behavioral Segmentation**
 - * Leveraged Customer Data Platform (CDP) data containing rich behavioral features — flight frequency, booking lead time, trip purpose (business/leisure), annual trip count, and spending patterns — to identify high-propensity hotel bookers.
 - * Applied K-Means clustering with PySpark on the existing hotel booker population to discover 10 distinct behavioral segments, capturing diverse customer profiles and booking patterns.
 - * Engineered a two-stage candidate selection mechanism: applied Euclidean distance thresholding to filter non-bookers near confirmed booker clusters, followed by cosine similarity ranking to score and prioritize users by behavioral similarity.
 - * Drove measurable business impact with a **30% increase** in inactive-to-active user conversion and a **20% uplift** in click-to-open rate (CTOR) of targeted hotel promotional emails and push notifications.
 - **LLM-Based Cargo Commodity Code Classification & Revenue Leakage Detection**
 - * Identified that ~20% of cargo shipment records carried mislabelled commodity codes causing revenue leakage; designed and delivered an end-to-end LLM-based pipeline to automatically detect and correct misclassifications.
 - * Built a structured prompt using priority tiers, confidence-based multi-label output, and domain-specific classification rules, leveraging commodity code definitions and historical patterns as reference context.
 - * Implemented a batched inference strategy (10 descriptions per LLM call) using LLaMA 4 Maverick via Databricks `ai_query()`, reducing API call overhead while maintaining classification accuracy of ~95%.
 - * Developed a downstream rate calculation engine applying weight slab logic, commodity-specific rate overrides, and min/max price computation against freight market rates — directly quantifying revenue impact of each mislabelled shipment.
 - * Persisted structured prediction results and freight market variance to Snowflake via REST API, powering a Power BI dashboard for business and revenue teams to monitor commodity mismatches and pricing discrepancies.
 - **ASK_PRISM — Conversational Text-to-SQL Intelligence System for Business Self-Serve Analytics**
 - * Designed and developed ASK_PRISM, an enterprise-wide conversational AI system enabling business users to query databases in natural language without requiring SQL expertise or BAU team intervention.
 - * Initiated deployment with ASK_PRISM_CDP on Databricks Genie Spaces, targeting CDP data for customer and behavioral analytics; successfully deployed to production.
 - * Architected an expansion onto Snowflake's Cortex layer — leveraging Cortex Analyst, AI Agents, and Snowflake Intelligence — to cover enterprise-wide data domains including Sales, Air Operations (AIROPS), India Data Hub (IDH), and Catering Operations (COPS).
 - * Configured and curated semantic models, table definitions, and domain-specific context for each data domain, enabling accurate natural language to SQL translation across diverse and complex schemas.
 - * Collaborated with a cross-functional team to democratize data access across multiple business units, significantly reducing query turnaround time that previously required manual SQL intervention.

PROJECTS

- **Air Pollutant Prediction using Sensor Data (CS771) | Guide: Prof. Purushottam Kar**

(Aug 2022 – Nov 2022)

- Built a linear regression model to predict O3 and NO2 concentrations using voltage readings, achieving O3 MAE of 5.63 and NO2 MAE of 6.3.

- Developed non-linear models incorporating temperature, humidity, and timestamps; KNN outperformed all others with O3 MAE of 3.1 and NO2 MAE of 1.6.
- **Sampling and Reconstruction of Volume Data (CS661) | Guide: Prof. Soumya Dutta** (Jan 2023 – Apr 2023)
 - Implemented Simple Random Sampling to sample a given percentage of points from the Isabel 3D dataset and performed reconstruction using nearest-neighbor and linear interpolation methods.
 - Attained a signal-to-noise ratio (SNR) of up to 40 across various sampling percentages.

ACHIEVEMENTS AND CERTIFICATIONS

- Secured **All India Rank 185** in **GATE CS 2022** amongst 77,257 candidates
- **Algorithmic Toolbox** — University of California San Diego (Coursera)
- **Career Essentials in Generative AI** — Microsoft and LinkedIn

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, C++, C
- **ML / Data Science:** Scikit-learn, NumPy, Pandas, PySpark, Matplotlib, Seaborn, Hugging Face Transformers, MLflow, REST APIs / FastAPI
- **Data Engineering:** Snowflake, Databricks, Delta Lake, Delta Live Tables, Apache Spark, Git, AWS, Flask, ~~ETL~~ ~~ETL~~
- **AI / LLM & Generative AI:** Large Language Models (LLMs), Prompt Engineering, Few-Shot Learning, NLP, Sentiment Analysis, Agentic AI, Multi-Agent Frameworks, Snowflake Cortex, Databricks Genie Spaces, Text-to-SQL, Power BI